A2.1 Origins of cells (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **STATE** two features of the early Earth's atmosphere that differ from today's. [A2.1.1] [2]

2 **LIST** the four requirements for the evolution of the first cells. [A2.1.3] [2]

3 **DEFINE** ribozyme. [A2.1.6] [1]

4 **STATE** what LUCA stands for, and roughly when it lived. [A2.1.7] [2]

5 **STATE** two likely features of LUCA's metabolism. [A2.1.9] [2]

Section B: Understanding

15 MARKS

6 **EXPLAIN** why a cell is the smallest unit of self-sustaining life, while a virus is not considered living. [A2.1.2] [3]

7 **EXPLAIN** how fatty acids could form vesicles without enzymes, and why this was important. [A2.1.5] [3]

8 **EXPLAIN** why RNA is thought to have been the first genetic material. [A2.1.6] [3]

9 **OUTLINE** how the universal genetic code suggests that all life shares one ancestor. [A2.1.7] [2]

10 **DESCRIBE** how a molecular clock is used to estimate when two groups shared an ancestor. [A2.1.8] [2]

11 **SUGGEST** why the first cells left no fossils. [A2.1.3] [2]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [A2.1.4]

A student repeats a Miller–Urey type experiment with three gas mixtures, each sparked for seven days.

GAS MIXTURE	DIFFERENT AMINO ACIDS DETECTED	TOTAL YIELD / MG
CH_4, NH_3, H_2, H_2O	11	42
CO_2, N_2, H_2O	3	4
CO_2, N_2, H_2O, IRON SULFIDE	6	15

a CALCULATE how many times greater the yield was with the first mixture than with the second. [1]

b COMPARE the results for the three mixtures. [2]

c Many scientists now think the early atmosphere was rich in carbon dioxide. DISCUSS what these results suggest about the original Miller–Urey conclusion. [3]

d SUGGEST why iron sulfide was added, with reference to hydrothermal vents. [2]

e STATE one variable that must be controlled when comparing the mixtures. [1]

f OUTLINE why these results do not show how a cell formed. [1]

Section D: Longer answer

6 MARKS

13 DISCUSS the evidence that life began near deep-sea hydrothermal vents. [A2.1.7, A2.1.8, A2.1.9] [6]